

HOTAN, China

WMO: 518280

Lat: 37.133N

Lon: 79.933E

Elev: 1375

StdP: 85.86

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-11.9	-9.8	-20.2	0.7	-2.9	-18.4	0.9	-1.4	4.5	-0.5	4.2	-1.1	1.4	250	0.408

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.9	35.4	17.6	34.0	17.3	32.6	16.9	20.1	30.8	19.2	30.1	18.4	29.3	2.4	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.5	13.9	25.3	15.4	13.0	24.1	14.6	12.3	23.7	64.3	30.8	60.9	30.3	58.2	29.4	24.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
5.5	4.7	4.2	DB	-13.0	38.1	3.4	1.2	-15.4	39.0	-17.4	39.6	-19.3	40.3	-21.8	41.1	
			WB	-13.7	21.7	3.2	1.3	-16.0	22.6	-17.8	23.4	-19.6	24.2	-21.9	25.2	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	14.1	-3.4	2.1	10.9	17.7	21.9	25.1	26.7	25.7	21.5	15.0	6.8	-1.1
	DBStd	10.98	4.07	4.18	4.37	4.08	3.37	2.98	3.01	2.93	2.85	3.46	4.04	3.52
	HDD10.0	1141	415	222	42	3	0	0	0	0	3	111	345	
	HDD18.3	2498	673	454	231	57	11	1	0	7	114	347	603	
	CDD10.0	2646	0	1	71	235	368	453	518	486	344	157	14	0
	CDD18.3	961	0	0	1	38	120	203	259	228	101	10	0	0
	CDH23.3	9227	0	0	16	309	1010	2014	2867	2251	700	59	0	0
	CDH26.7	3729	0	0	2	73	326	837	1345	951	191	4	0	0
Wind	WSAvg	2.1	1.6	1.8	2.1	2.3	2.4	2.5	2.3	2.2	2.1	1.9	1.8	1.6
Precipitation	PrecAvg	38	1	2	2	2	7	8	6	4	2	1	0	1
	PrecMax	113	11	17	17	22	42	33	25	22	22	14	8	18
	PrecMin	3	0	0	0	0	0	0	0	0	0	0	0	0
	PrecStd	24	3	4	4	4	9	9	7	5	5	3	1	3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.4	15.7	25.2	31.2	33.5	36.3	37.6	36.6	32.4	27.0	20.4	12.1
		MCWB	2.2	4.8	9.2	13.0	14.9	16.6	17.9	18.2	15.8	12.5	7.8	2.8
	2%	DB	6.9	12.4	22.5	28.7	31.6	34.1	35.9	34.6	30.5	25.0	17.4	8.4
		MCWB	-0.2	3.3	8.1	11.9	14.0	16.2	18.0	17.8	15.4	11.8	6.4	1.0
	5%	DB	4.1	10.2	20.5	26.8	30.0	32.7	34.4	33.0	28.9	23.3	15.1	5.9
		MCWB	-1.6	2.2	7.4	11.1	13.8	16.1	17.8	17.8	14.9	11.2	5.3	-0.2
	10%	DB	2.1	8.3	18.3	24.9	28.5	31.2	32.8	31.5	27.3	21.3	13.1	3.9
		MCWB	-2.7	1.2	6.5	10.3	13.4	15.9	17.6	17.6	14.6	10.1	4.4	-1.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.2	5.4	9.7	13.7	16.6	19.3	21.8	21.4	18.3	13.6	8.3	3.1
		MCDB	11.0	13.8	23.2	28.9	29.6	31.1	32.0	31.6	28.8	24.5	19.4	11.2
	2%	WB	-0.1	3.8	8.5	12.6	15.5	18.1	20.4	20.0	16.9	12.5	6.7	1.2
		MCDB	6.5	11.4	21.1	27.0	28.4	29.8	31.0	30.6	26.7	23.3	16.7	8.0
	5%	WB	-1.5	2.6	7.6	11.7	14.7	17.3	19.4	19.1	16.1	11.6	5.5	-0.1
		MCDB	3.9	9.5	19.5	24.7	27.1	29.0	30.4	29.8	26.0	22.0	14.5	5.2
	10%	WB	-2.7	1.5	6.7	10.9	14.1	16.7	18.7	18.3	15.3	10.7	4.6	-1.2
		MCDB	1.7	7.9	17.6	23.2	26.0	28.2	29.6	29.1	24.9	20.5	12.6	3.5
Mean Daily Temperature Range	5% DB	MDBR	7.8	8.4	9.9	10.5	10.6	10.1	9.9	9.6	10.2	11.4	9.9	7.8
		MCDBR	10.4	10.6	12.4	13.3	12.7	12.1	12.3	12.1	12.3	12.5	12.2	10.4
		MCWBR	6.2	5.7	5.8	5.9	5.4	4.9	5.4	5.7	5.6	5.6	6.2	6.1
	5% WB	MCDBR	10.0	10.3	11.7	11.6	11.0	10.6	10.7	10.8	10.9	11.7	11.7	9.9
		MCWBR	6.1	5.5	5.5	5.2	5.0	4.7	5.5	5.6	5.5	5.4	6.1	5.9
Clear-Sky Solar Irradiance	taub	0.332	0.401	0.523	0.615	0.641	0.656	0.645	0.680	0.622	0.477	0.387	0.349	
	taud	2.132	1.946	1.654	1.506	1.507	1.525	1.566	1.502	1.568	1.790	1.984	2.072	
	Ebn at Noon	830	804	732	694	687	676	679	639	643	709	746	772	
	Edh at Noon	114	157	232	285	289	284	271	283	251	182	132	114	
All-Sky Solar Radiation	RadAvg	2.83	3.71	5.12	6.05	6.75	6.72	6.32	5.75	5.28	4.61	3.40	2.72	
	RadStd	0.35	0.27	0.31	0.23	0.32	0.33	0.49	0.50	0.30	0.13	0.17	0.18	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.57	N/A	N/A	N/A	-0.96	-1.28	N/A	-140	+151	+61
(47)	Regional (2 neighbors)	+0.38	N/A	N/A	N/A	-0.88	-0.80	N/A	-140	+151	+61

Nomenclature: See separate page